

Tourism Demand and Spatial Dependence in Italian Municipalities

Andrea Battaglia

Link Github: https://github.com/01andreabattaglia/Geospatial_Analysis

Abstract

This paper investigates the spatial dimension of tourism demand across Italian municipalities. It examines the relationship between overnight stays and municipal endowments of cultural, natural, tourism, recreational and accessibility resources, while accounting for spatial interactions between neighbouring municipalities. Municipality-level data from ISTAT, OpenStreetMap and UNESCO are combined into a cross-sectional dataset covering the Italian territory. The analysis applies exploratory spatial data analysis to identify spatial patterns and clusters of tourism demand, followed by spatial econometric models to account for spatial dependence and estimate both local and spillover effects. Particular attention is given to the Spatial Durbin Model and to whether spillovers generated by tourism resources reflect complementarity or competition. The study provides an integrated assessment of the determinants and spatial structure of tourism demand at the municipal level, highlighting the importance of considering the wider territorial context of tourism destinations.

1 Research question

1.1 Motivation

Tourism demand is spatially structured rather than randomly distributed. Municipalities often share cultural, natural, infrastructural and tourism resources with neighbouring areas, while tourists may visit several destinations during the same trip. Consequently, a municipality's tourism performance may depend not only on its own characteristics but also on those of surrounding municipalities.

This spatial dimension is particularly relevant in Italy, where coastal, mountain, historic, rural and lake destinations form interconnected tourism systems with substantial differences in their resource endowments and accessibility. Building on Tobler's first law of geography, which states that nearby locations tend to be more closely related than distant ones (Tobler, 1970), this study examines both the relationship between municipal tourism resources and overnight stays and the extent to which tourism demand and resources generate spatial spillovers. The analysis also considers whether these interactions reflect complementarity between neighbouring destinations or competition for tourism demand.

1.2 Literature review

The research question builds on a well-established strand of spatial econometric literature applied to tourism demand. The theoretical foundation for the entire analysis comes from Tobler (1970), who established that geographical proximity is a systematic source of statistical dependence, so that standard regression techniques, which assume independent observations, are generally inappropriate for geo-referenced tourism data. Building on this foundation, Yang and Wong (2012) offer direct empirical evidence that tourism flows are subject to spatial spillovers, showing through a spatial econometric approach that the attractiveness of one destination affects tourism outcomes in neighbouring destinations, and that ignoring this interdependence leads to biased estimates of the determinants of tourism demand. Yang and Fik (2014) extend this line of research by showing that regional tourism growth itself has a distinct spatial component, reinforcing the idea that municipal-level tourism performance cannot be fully understood by looking at local characteristics alone.

Evidence specific to the Italian context comes from Zamparini et al. (2017), which show that local tourism demand in Italian provinces depends on a combination of economic and non-economic characteristics of the destinations, many of which (accommodation capacity, cultural heritage, accessibility) correspond directly to the explanatory variables used in this project. In a related vein, Pompili et al. (2019) model international tourist choices across Italian provinces within a joint demand-supply framework that explicitly incorporates spatial effects, providing methodological support for treating spatial dependence as a structural feature of the data-generating process. Closer still to the empirical strategy adopted here, Scotti et al. (2024) estimate a log-log gravity model on mobile-network data for Lombardy municipalities, showing that accommodation capacity and cultural and natural endowments raise overnight-stay flows while entertainment and transport facilities instead favour same-day visits; this offers direct support, at the sub-provincial Italian scale, for the multiplicative, log-log functional form adopted for the resource variables used in this project. Taken together, these contributions motivate both the substantive expectation that Italian municipal tourism demand is spatially dependent, and the methodological choice of spatial econometric models, rather than a standard non-spatial regression, as the appropriate empirical strategy.

A further contribution grounds the choice of explana-

tory variables used to characterise the tourism endowment of each municipality. Dwyer and Kim (2003) develop an integrated model of destination competitiveness in which tourism performance is explained by a set of core resources and attractors (natural, cultural, and heritage resources), supporting factors and infrastructure (accommodation capacity, transport, general infrastructure) and destination management. This taxonomy provides a conceptual justification, independent of the spatial dimension of the analysis, for considering these categories as the relevant determining factors of municipal tourism demand.

At the municipal level, the relevance of spatial interactions is particularly salient because neighbouring municipalities can either compete for the same visitors or benefit from tourism complementarity. On the one hand, a concentration of attractions within a territory may encourage tourists to visit multiple municipalities during the same trip, generating positive spillovers. On the other hand, nearby destinations may offer substitute tourism products and therefore compete for the same pool of visitors. The empirical analysis does not assume a priori that spatial spillovers are necessarily positive, and is designed to let the data discriminate between these two mechanisms.

1.3 Research questions and geographical scope

The general research question is operationalised into three specific sub-questions, investigated across the Italian territory at the municipal level.

RQ1. *Is tourism demand, measured through overnight stays, spatially autocorrelated across Italian municipalities, and are there statistically significant geographical clusters of municipalities with high or low tourism demand?*

RQ2. *Which municipal endowments of cultural, natural, entertainment and accessibility resources are associated with tourism demand, once spatial dependence in the data is accounted for?*

RQ3. *Does the tourism endowment of a municipality generate spillover effects on tourism demand in neighbouring municipalities, and are these effects complementary or competitive?*

Each observation corresponds to an Italian municipality. The municipality is used as the unit of analysis because it allows administrative, tourism and geospatial information to be combined, and provides the basis for constructing a spatial weights matrix describing geographical relationships between municipalities.

The national scale is particularly suitable for these research questions because Italy displays substantial geographical heterogeneity in tourism resources and destination characteristics. Coastal, mountain, historic, island, lake and inland municipalities differ considerably in terms of natural and cultural resources, accommodation capacity, accessibility and tourism activities. This diversity provides an appropriate setting for examining both the spatial distribution of tourism demand and

the extent to which municipal characteristics and neighbouring destinations influence tourism performance.

1.4 Research hypotheses

The empirical analysis is guided by four research hypotheses, each corresponding to one of the research questions introduced above.

H₁: Tourism demand is positively spatially autocorrelated across Italian municipalities. Municipalities with high levels of overnight stays tend to be located close to other municipalities with high tourism demand, while low-demand municipalities tend to cluster together.

H₂: Municipalities with a greater endowment of tourism resources and infrastructure exhibit higher tourism demand, measured through overnight stays.

H₃: Tourism resources generate spatial spillovers that extend beyond the boundaries of the municipality in which they are located.

H₄: The sign of spatial spillovers is not necessarily homogeneous across resources: some resources are expected to generate positive complementarity effects, while others may generate negative competitive effects.

These hypotheses are evaluated empirically in the remainder of the report through exploratory spatial data analysis (RQ1) and spatial econometric modelling (RQ2 and RQ3).

2 Description of data

The empirical analysis combines administrative, spatial and open geospatial data to construct a municipality-level dataset for Italian municipalities. The unit of observation is the municipality, identified through the ISTAT municipal boundaries. The final dataset combines information on tourism demand with measures of cultural, natural, tourism, entertainment and accessibility resources.

The main sources are the Italian National Institute of Statistics (ISTAT), OpenStreetMap (OSM) and the UNESCO list of Italian World Heritage Sites. OSM is particularly important because it allows the construction of a detailed set of municipality-level indicators describing tourism resources and local infrastructure. The data are spatially aggregated to municipal boundaries and subsequently merged into a single cross-sectional dataset.

2.1 Data sources

The datasets used in the analysis can be grouped into three main sources.

ISTAT. ISTAT provides the official municipal boundaries, territorial classifications and tourism accommodation data. These data are used for the geographical definition of the units of analysis, the classification of municipalities by altitude and island status and the measurement of accommodation capacity.

OpenStreetMap. OSM provides georeferenced information on cultural attractions, architectural heritage, natural resources, recreational activities, entertainment facilities and local transport infrastructure. The data were extracted through Overpass queries using specific OSM tags and subsequently assigned to municipalities through spatial operations.

UNESCO. The official list of UNESCO World Heritage Sites in Italy is used to construct the number of UNESCO sites located within each municipality.

The use of OpenStreetMap requires particular attention because it is a collaborative and continuously updated geographic database. Consequently, OSM-based variables should be interpreted as measures of the resources recorded in OSM rather than as official administrative inventories. Differences in mapping intensity across municipalities may therefore introduce measurement differences that are not related to actual differences in tourism endowment.

2.2 Municipality boundaries and spatial framework

The geographical framework is based on the ISTAT municipal boundary shapefile `Com01012024_g_WGS84.shp`, referring to the municipal configuration of 1 January 2024. The shapefile provides the polygons used as the spatial units of analysis and as the basis for the spatial aggregation of OSM information.

Geometries are initially provided in WGS84 and are reprojected into a metric coordinate reference system for operations involving distances, lengths, and areas. This step is necessary because calculations such as kilometres of coastline or square kilometres of protected areas should not be performed directly in geographic coordinates.

2.3 Variables and data construction

The explanatory variables are organised into five groups: cultural and heritage resources, natural resources, tourism infrastructure, recreational and entertainment activities, and supporting factors.

2.3.1 Cultural and heritage resources

The first group measures the cultural attractiveness of municipalities.

Historical sites and museums. Number of museums, galleries, monuments, memorials, ruins, archaeological sites and forts recorded in OSM.

Architectural and artistic features. Number of churches, castles, manors, palaces, city gates and artworks recorded in OSM.

UNESCO sites. Number of UNESCO World Heritage Sites located within the municipality. This variable is obtained by spatially joining the UNESCO geographical information with the ISTAT municipal boundaries.

For the two OSM-based cultural variables, nodes, ways and relations are included in the extraction. For ways and relations, the centre coordinates supplied by Overpass are used to assign the feature to a municipality. This approach provides a consistent municipality-level count while avoiding the need to assign an entire polygon or line feature to multiple municipalities.

2.3.2 Natural resources

Natural tourism endowment is captured through both geographical characteristics and protected areas.

Coastline length. Kilometres of coastline falling within the municipality. Coastline geometries are extracted from OSM and intersected with municipal boundaries.

Lake shoreline length. Kilometres of lake shoreline falling within the municipality. Lake geometries are obtained from OSM and intersected with municipal boundaries.

Island municipality. Binary variable equal to one when the municipality is classified by ISTAT as an island municipality and zero otherwise.

Altitude category (ZONA_ALT). ISTAT classification of the national territory into homogeneous altimetric zones, obtained by aggregating contiguous municipalities on the basis of altitude threshold values. The variable takes five categories: inland mountain areas (1), coastal mountain areas (2), inland hill areas (3), coastal hill areas (4), and plain areas (5).

Protected areas. Total surface of protected or natural areas located within the municipality, measured in square kilometres. OSM protected-area geometries are extracted and spatially intersected with municipal boundaries; the area of the resulting intersections is calculated in a metric CRS and expressed in km². Municipalities without mapped protected areas are assigned a value of zero.

The construction of the coastline, lake shoreline and protected-area variables requires geometric rather than point-based aggregation. In these cases, the OSM geometries are reprojected into a metric CRS before calculating lengths or areas.

The procedure used for the three spatial variables is illustrated in Figures 1, 2, and 3.

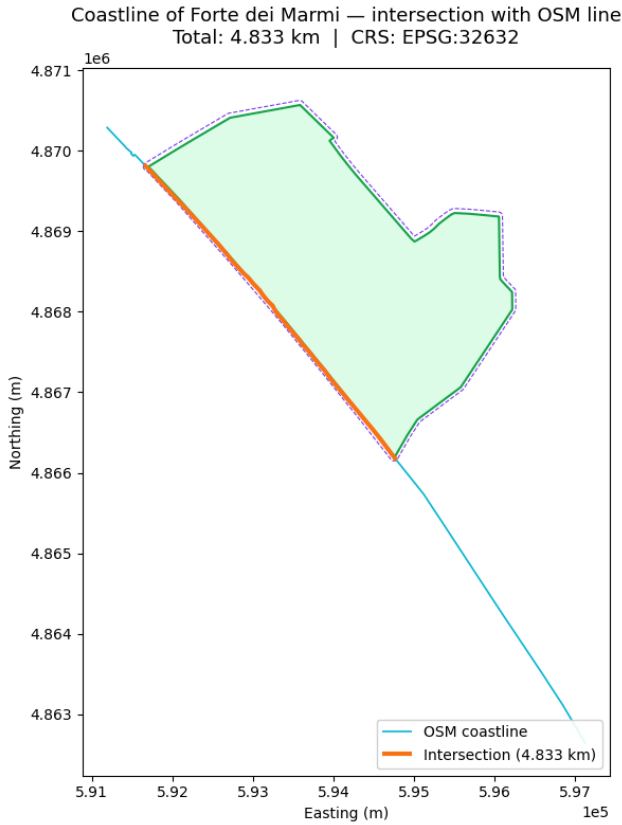


Figure 1: Coastline: Forte dei Marmi.

Lake shoreline of Como — intersection with OSM perimeter
Total: 8.032 km | CRS: EPSG:32632

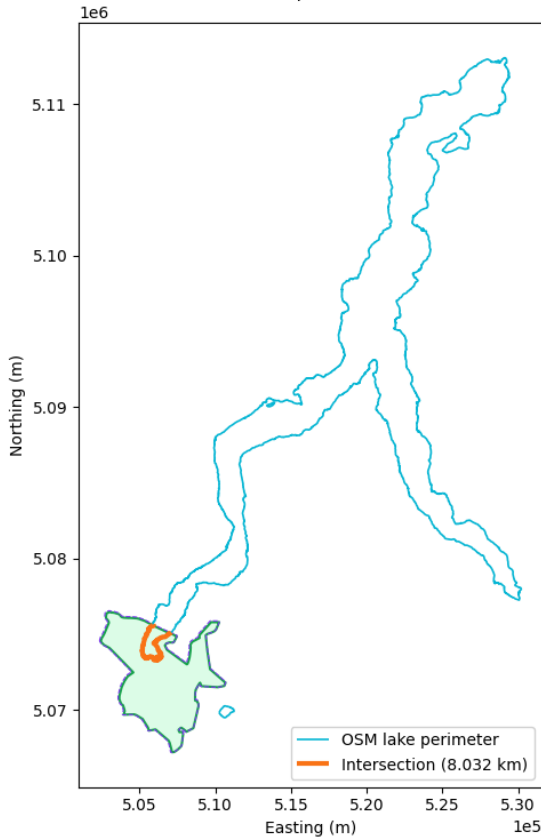


Figure 2: Lake shoreline: Como.

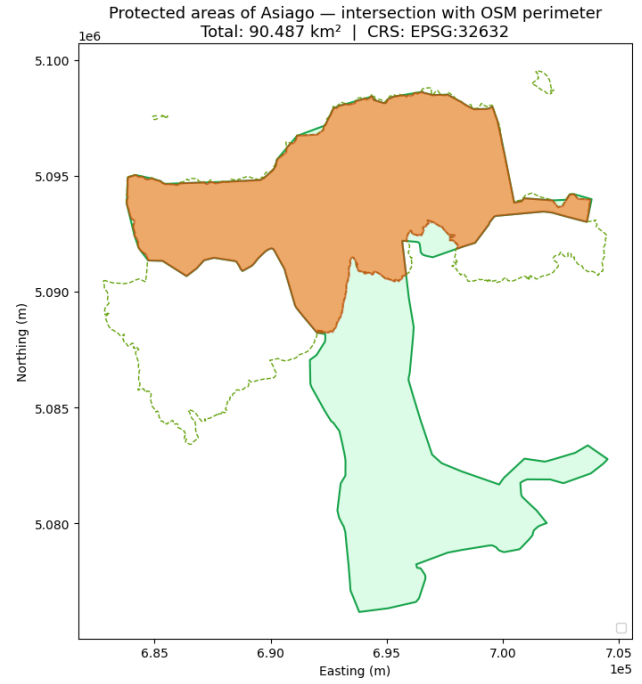


Figure 3: Protected areas: Asiago.

2.3.3 Tourism infrastructure

Tourism accommodation capacity is obtained from IS-TAT tourism statistics and is divided into hotel and extra-hotel accommodation.

Hotel beds. Total number of beds in hotels and tourist residences, including 1–5 star hotels, 5-star deluxe establishments and tourist residences (RTA).

Extra-hotel beds. Total number of beds in non-hotel accommodation, including campsites and tourist villages, holiday dwellings, farmhouses, youth hostels, tourist dormitories, mountain huts, bed and breakfasts, and other accommodation establishments.

The accommodation variables measure the capacity of the tourism supply rather than realised tourism demand. This distinction is important because accommodation capacity is used as an explanatory variable for overnight stays in the subsequent analysis.

2.3.4 Recreational and tourism activities

A further set of OSM variables captures the range of activities available to visitors.

Sports facilities. Number of sports facilities and tourism-related sports infrastructure recorded in OSM. The extraction includes sports centres, stadiums, sports pitches, swimming pools, sports halls, marinas, slipways, sailing and surfing facilities, scuba-diving facilities, ski slopes, cable cars, chair lifts, golf facilities, and tennis facilities.

Nature-based activities. Number of nature-based activity facilities and tourism-related outdoor in-

infrastructure recorded in OSM. The extraction includes hiking and cycling routes, nature reserves, alpine and wilderness huts, campsites, and spas.

Amusement and theme parks. Number of amusement and theme park facilities and tourism-related recreational infrastructure recorded in OSM. The extraction includes theme parks and water parks.

Nightlife. Number of nightlife facilities and tourism-related entertainment infrastructure recorded in OSM. The extraction includes bars, pubs and nightclubs.

These variables are intended to capture the diversity of tourism-related activities available in a municipality. They complement the more traditional measures of cultural heritage and accommodation capacity and are therefore particularly relevant for testing whether different forms of tourism endowment have different effects on tourism demand.

2.3.5 Supporting factors and accessibility

The final group contains variables describing local accessibility.

Local transport systems. Number of public transport stops and stations, including bus stops, railway stations, railway halts, and public-transport stop positions. These features are extracted from OSM.

Airport accessibility. Straight-line distance from the municipality centroid to the nearest airport, measured in kilometres. Municipal centroids and airport locations are projected into a metric CRS, and a nearest-neighbour spatial join is used to identify the closest airport for each municipality. The resulting variable, `airport_straight_km`, measures geographical proximity to airport infrastructure rather than actual travel time or road accessibility.

2.4 Tourism demand

The dependent variable is tourism demand, measured by the number of overnight stays in accommodation establishments. The measure is obtained from ISTAT tourism statistics and is defined at the municipal level.

Overnight stays are preferred to tourist arrivals as the principal measure of demand because they capture both the number of visitors and the duration of their stay. A municipality receiving a smaller number of tourists who remain for several nights can therefore be distinguished from a municipality characterised primarily by short visits.

The distribution of overnight stays is expected to be highly heterogeneous across Italian municipalities, with a relatively small number of municipalities accounting for a substantial proportion of total tourism demand. This characteristic motivates the use of exploratory spatial data analysis in the first empirical stage and, where appropriate, transformations of highly skewed variables in the econometric analysis.

2.5 Data cleaning and wrangling

The construction of the final dataset involves several data-cleaning and spatial-wrangling steps.

First, all datasets are harmonised to the municipality identifier used by ISTAT. Spatial joins are preferred for georeferenced information.

Second, OSM data are filtered according to the specific tags defining each conceptual category. The extraction includes the relevant node, way and relation objects where appropriate.

Third, geometries are checked for missing or invalid values. Line and polygon geometries are converted to the appropriate geometric representation when necessary: polygon boundaries are used for coastline and lake-shoreline calculations, while polygon intersections are used for protected-area calculations.

Fourth, all distance, length and area calculations are performed after reprojection into a metric coordinate reference system. Coastline and lake-shoreline measures are then expressed in kilometres, while protected areas are expressed in square kilometres and subsequently converted into a percentage of the municipality's total area.

Fifth, OSM point-based features are assigned to municipalities and aggregated by counting the number of relevant features in each municipality. For variables based on spatial intersection, only the portion of the resource falling within the municipal boundary is retained.

Finally, the separate datasets are merged into a single municipality-level analytical dataset.

2.6 Data restrictions and limitations

The datasets used in the analysis are subject to different restrictions and limitations.

ISTAT data are official statistical data and are used according to the conditions established by the relevant ISTAT data products. The municipal boundary dataset is used as the geographical reference for the analysis.

OpenStreetMap data are openly available but are produced through collaborative mapping. Their completeness and positional accuracy can vary geographically and across types of features. The resulting OSM indicators should therefore be interpreted as measures of mapped tourism resources rather than complete official inventories.

The UNESCO data identify World Heritage Sites but do not necessarily capture the full cultural or historical endowment of a municipality. They therefore complement, rather than replace, the broader OSM-based cultural indicators.

A further limitation concerns the temporal dimension of the datasets. OSM is continuously updated, whereas ISTAT statistics and municipal boundaries refer to specific observation periods. The analysis therefore requires the different sources to be harmonised as closely as possible in time. Any remaining temporal mismatch is considered when interpreting the empirical results.

Despite these limitations, the combination of official statistics and detailed geospatial information provides a sufficiently rich database for investigating both the determinants of municipal tourism demand and its spatial dependence across Italian municipalities.

3 Results

The empirical analysis is organised around the three specific research questions introduced above: spatial autocorrelation in tourism demand (RQ1), the resource determinants of tourism demand once spatial dependence is accounted for (RQ2), and the spillover effects that municipal resources generate on neighbouring municipalities (RQ3).

3.1 RQ1: Spatial autocorrelation in tourism demand

3.1.1 Global spatial autocorrelation

The first stage of the empirical analysis addresses RQ1: whether tourism demand is spatially autocorrelated across Italian municipalities. Tourism demand is measured through overnight stays, transformed as $\log(1 + \text{overnight stays})$ to reduce the influence of extreme observations. Spatial relationships are represented through a queen-contiguity weights matrix, row-standardised, in which municipalities sharing a boundary edge or vertex are treated as neighbours; a small number of non-contiguous municipalities are linked to their nearest neighbour to avoid isolated units.

The global Moran's I statistic tests whether municipalities with similar tourism demand are geographically concentrated. The estimated value is 0.5156, and both the randomisation test and a Monte Carlo test based on 999 permutations reject the null hypothesis of spatial randomness ($p < 0.001$ and $p = 0.001$, respectively). This magnitude indicates a substantial degree of positive spatial association: high-demand municipalities tend to be surrounded by other high-demand municipalities, and vice versa. The result offers initial support for H_1 and confirms that tourism demand cannot be treated as spatially independent across municipalities.

This pattern is consistent with the geography of Italian tourism, where destinations rarely function as isolated administrative units but form broader systems built on shared natural resources, cultural landscapes, transport links and accommodation capacity. Figure 4 illustrates the relationship graphically: municipal tourism demand on the horizontal axis against its spatial lag on the vertical axis, with the concentration of points in the High-High and Low-Low quadrants mirroring the global statistic. Since Moran's I does not identify *where* this dependence occurs, the analysis proceeds to a local decomposition.

3.1.2 Local spatial clusters

Local Indicators of Spatial Association (LISA, $\alpha = 0.05$) classify each municipality according to its relationship

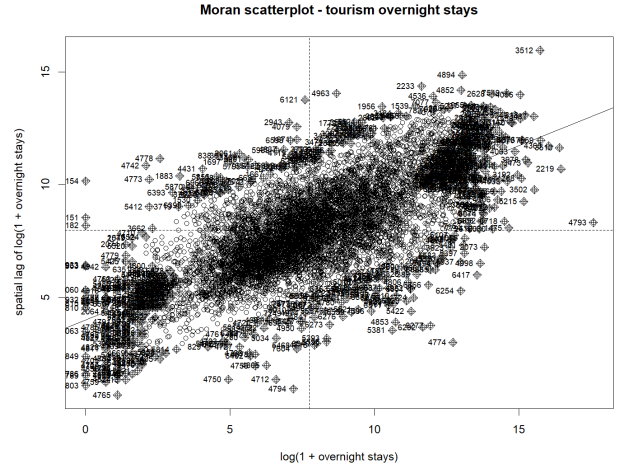


Figure 4: Moran scatterplot of tourism overnight stays

with neighbouring tourism demand: High-High, Low-Low, High-Low, Low-High, or Not significant. Table 1 reports the results: 1,093 municipalities fall into the High-High category and 644 into Low-Low, together accounting for 1,737 municipalities in statistically significant clusters of similar demand. A smaller set of 96 High-Low and 65 Low-High municipalities represent spatial outliers, while the remaining 6,001 municipalities show no significant local association.

Table 1: LISA classification of Italian municipalities

Cluster	Number of municipalities
High-High	1,093
Low-Low	644
High-Low	96
Low-High	65
Not significant	6,001
Total	7,899

Figure 5 maps these clusters. Moving from north to south, High-High municipalities trace out a few recognisable Italian tourism systems: the Alpine area around the Dolomites, where a shared mountain landscape and ski infrastructure integrate neighbouring municipalities; Lake Garda, where visitors typically move across several lakeside municipalities within the same trip; the northern Adriatic coast across Veneto, Emilia-Romagna and Marche, a continuous strip of beach-tourism accommodation and accessibility; central Tuscany and Umbria, reflecting a dense and contiguous supply of cultural, historical and landscape resources; the coastal municipalities of Sardinia, particularly along the northern and southern shorelines; parts of Puglia, combining coastal and cultural attractions; and the coastal areas of Sicily, where beach tourism coexists with historical and archaeological resources.

Low-Low clusters are equally informative, identifying territories where limited tourism demand is shared across neighbouring municipalities rather than confined to individual units. This pattern is particularly visible

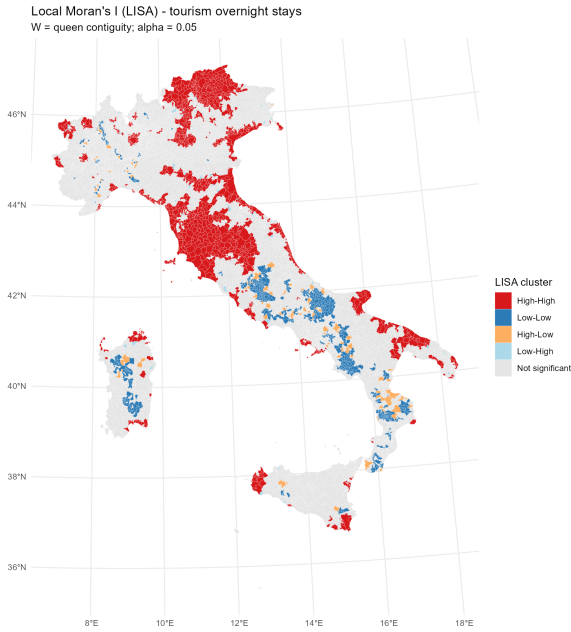


Figure 5: Local Moran's I (LISA) clusters of tourism overnight stays

in the inland Apennine areas of Abruzzo, Lazio, Campania and Calabria, and in the interior of Sardinia, areas typically characterised by lower accessibility, thinner accommodation supply, and a weaker consolidated destination identity relative to the coastal and Alpine systems described previously. The High-Low and Low-High outliers, by contrast, show that proximity to a successful destination does not automatically translate into comparable performance, suggesting that municipal-level characteristics can still differentiate outcomes within a given territorial context.

Taken together, the global and local statistics confirm H_1 : tourism demand is positively and significantly spatially autocorrelated across Italian municipalities, with high-demand and low-demand municipalities each tending to cluster geographically.

3.2 RQ2: Determinants of tourism demand

RQ2 asks which municipal endowments are associated with tourism demand once spatial dependence is accounted for. The results indicate that accommodation capacity is the strongest positive determinant of tourism demand, while recreational, natural and accessibility resources also matter. At the same time, several resources generate different effects within the municipality and across neighbouring municipalities, confirming that tourism demand is shaped by the wider spatial context.

The OLS residuals show strong spatial dependence, with Moran's $I = 0.371$ ($p < 0.001$). The Lagrange multiplier tests also reject the absence of spatial dependence (RSerr = 3004, adjRSerr = 1654, RSlag = 1420.7, adjRSlag = 70.9; all $p < 0.001$), supporting the use of spatial econometric models rather than OLS alone.

3.2.1 Model selection

Table 2 compares the fit of the estimated spatial specifications. The SDM provides the best fit, with the lowest AIC (24431.35) and the highest log-likelihood (-12185.67). Likelihood ratio tests further confirm the superiority of the SDM over the SAR ($LR = 1070$, $p < 0.001$) and over the SEM ($LR = 148.3$, $p < 0.001$). The SDM is therefore used as the preferred model.

Table 2: Comparison of spatial specifications

Model	AIC	logLik	Parameters
OLS	26744.71	-13351.35	22
SLX	26510.05	-13226.03	33
SAR	25485.31	-12720.65	24
SEM	24563.64	-12259.82	24
SDEM	24504.10	-12222.05	34
SDM	24431.35	-12185.67	34

The SDM has a positive and highly significant spatial autoregressive parameter, $\rho = 0.594$ (standard error 0.011, $p < 0.001$), confirming substantial spatial dependence in tourism demand.

3.2.2 SDM coefficient estimates

Table 3 reports the full set of coefficient estimates from the SDM, including the spatial lags of the Durbin variables. The direct (own-municipality) effects are given by the coefficients on the untransformed variables.

At the municipal level, the coefficient estimates show that accommodation capacity (both hotel and non-hotel beds) exerts the strongest positive effect on local tourism demand. Sports facilities and nature-based resources also have significant positive direct effects, as do nightlife, transport points, coastline length (sea and lake), and altitude zone relative to the reference category. Protected area extent and airport distance are negatively associated with local demand, while museums, architecture, theme parks, island status and UNESCO status are not statistically significant direct predictors.

Overall, the evidence supports H_2 : municipalities endowed with greater tourism accommodation capacity, recreational resources, natural attractions and accessibility tend to exhibit higher tourism demand. The spatial lag terms reported in the lower panel of Table 3 are discussed together with the full direct/indirect/total impact decomposition in the following section, which examines RQ3 and the associated spillover mechanisms explicitly.

3.3 RQ3: Spatial Spillovers of Tourism Resources

To assess whether municipal tourism endowments generate spillover effects on neighbouring municipalities, we estimate the Spatial Durbin Model (SDM) selected in RQ2. The SDM decomposes total effects into *direct* (own-municipality) and *indirect* (neighbour-driven) impacts. Positive indirect effects indicate complementarity

Table 3: SDM coefficient estimates

Variable	Estimate	Pr(> z)	Variable	Estimate	Pr(> z)
(Intercept)	2.882	< 0.001	log_theme_parks	0.011	0.467
log_hotel_beds	0.911	< 0.001	log_nightlife	0.118	< 0.001
log_non_hotel_beds	0.765	< 0.001	log_transport_pts	0.097	< 0.001
island_municipalityIsland	0.097	0.630	log_airport_dist	-0.085	< 0.001
altitude_zone2	0.377	0.001	has_unescoYes	-0.116	0.469
altitude_zone3	0.190	< 0.001			
altitude_zone4	0.479	< 0.001	<i>Spatial lags (Durbin terms)</i>		
altitude_zone5	0.493	< 0.001	lag.log_hotel_beds	-0.399	< 0.001
log_sea_coast_km	0.127	< 0.001	lag.log_non_hotel_beds	-0.550	< 0.001
log_lake_coast_km	0.043	0.004	lag.log_sea_coast_km	-0.186	< 0.001
log_protected_area	-0.041	0.010	lag.log_lake_coast_km	0.023	0.422
log_museums	-0.017	0.253	lag.log_protected_area	-0.019	0.331
log_architecture	0.020	0.339	lag.log_sports	-0.234	< 0.001
log_sports	0.402	< 0.001	lag.log_nature	0.124	< 0.001
log_nature	0.068	0.002	lag.log_transport_pts	-0.134	< 0.001

(neighbours' resources boost local demand), while negative indirect effects indicate competition (substitution or diversion of tourists).

Figure 6 reports direct and indirect effects for all explanatory variables. Several resources exhibit significant spatial spillovers. *Hotel beds* show a positive indirect effect (0.33, $p < 0.001$), suggesting that nearby hotel capacity complements local tourism demand. In contrast, *non-hotel beds* have a negative indirect effect (-0.22 , $p = 0.003$), indicating competitive substitution with alternative accommodations.

Among natural amenities, *sea coast* exerts a negative spillover (-0.26 , $p < 0.001$), implying coastal municipalities compete for tourists, whereas *lake coast* shows a mildly positive spillover (0.11, $p = 0.071$). Most notably, *nature-based resources* have a strong positive indirect effect (0.38, $p < 0.001$), indicating that natural attractions in neighbouring municipalities enhance local tourism demand.

Accessibility variables, given by *public transport points* (-0.18 , $p < 0.001$) exhibits competitive spillovers, suggesting that better accessibility in neighbouring areas may divert tourism flows. Cultural resources (museums, architecture, UNESCO) and entertainment (theme parks, sports) do not show significant indirect effects, implying their spatial influence remains predominantly local.

To visualise the spillover channel, Figure 7 maps the WX component for nature-based resources, the contribution of neighbours' natural endowment to each municipality's demand, scaled by the estimated spatial lag coefficient ($\theta = 0.151$). The map reveals heterogeneous spatial patterns, with positive (red) and negative (blue) spillover intensities concentrated in different regions of Italy.

Overall, these findings support $H3$ (spillovers exist beyond municipal boundaries), as several indirect effects are statistically significant, and $H4$ (heterogeneous sign), because both complementary (hotel beds, nature, lake coast, nightlife) and competitive (non-hotel beds, sea coast, protected areas, transport, airport distance) spillovers are observed.

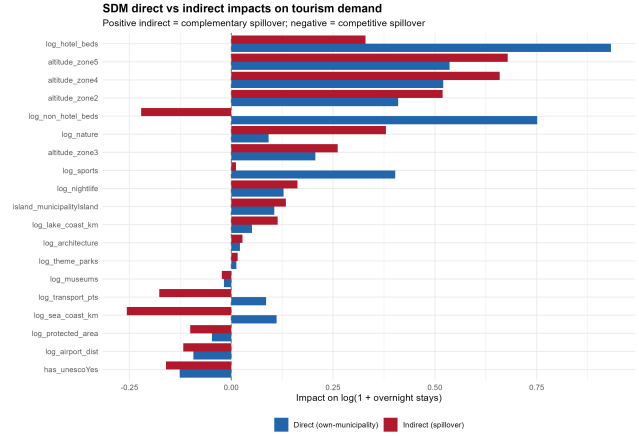


Figure 6: Direct vs indirect impacts of the SDM. Positive indirect effects indicate complementary spillovers; negative indicate competitive spillovers.

4 Conclusion

4.1 Summary of findings

This report examined the spatial dimension of tourism demand across Italian municipalities, addressing three related questions: whether tourism demand is spatially clustered, which municipal resources are associated with tourism demand once spatial dependence is taken into account, and whether these resources generate spillover effects on neighbouring municipalities.

The exploratory spatial analysis showed that tourism demand is not randomly distributed across the Italian territory but is organised into clearly identifiable geographical clusters, with high-demand municipalities concentrated in recognisable tourism systems (Alpine areas, lakeside districts, coastal strips, and historic-cultural regions) and low-demand municipalities clustering in less accessible inland areas. This confirms that municipal tourism performance cannot be properly understood in isolation from the surrounding territory.

The econometric analysis, based on a Spatial Durbin Model selected over alternative spatial specifications,

Spillover from neighbours' nature-based resource endowment
theta = 0.124 (complementary)

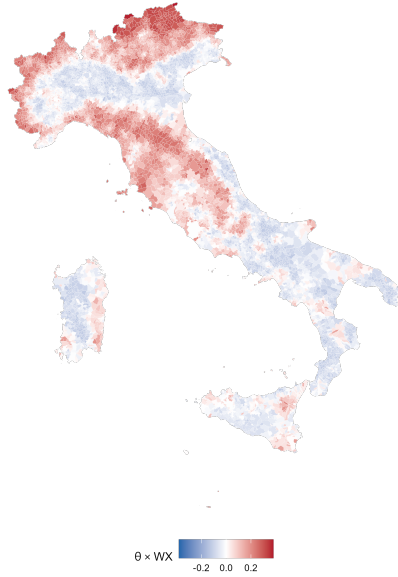


Figure 7: Spatial distribution of the spillover channel for nature-based resources ($\theta \cdot WX_{\text{nature}}$). Red (positive) areas benefit from neighbours' natural endowments; blue (negative) areas suffer from competition.

indicated that accommodation capacity is the strongest driver of local tourism demand, complemented by recreational, natural and accessibility resources, while some cultural variables did not show a significant direct association once spatial effects were controlled for. Beyond these local effects, several resources were also found to generate spillovers on neighbouring municipalities, and these spillovers were not uniformly positive: some resources appeared to benefit neighbouring destinations, consistent with complementarity within shared tourism systems, while others appeared to divert demand away from surrounding municipalities, consistent with competition for the same pool of visitors. Taken together, these results support the idea that tourism demand in Italy is shaped jointly by local endowments and by the broader spatial context in which municipalities are embedded, with both cooperative and competitive dynamics coexisting across the territory.

4.2 Limitations

Despite the richness of the dataset and the robustness checks carried out, three main limitations should be acknowledged when interpreting the results.

Reliance on OpenStreetMap data. A substantial share of the explanatory variables is derived from OpenStreetMap, a collaborative and continuously updated database whose mapping intensity is not homogeneous across the Italian territory. Municipalities that are more actively mapped may appear artificially better endowed with cultural, natural or recreational resources than municipalities where the same resources exist but have not been systematically recorded, introducing a measurement

bias that is difficult to fully separate from genuine differences in tourism endowment.

Imperfect model fit. Although the Spatial Durbin Model clearly outperforms the non-spatial and more restrictive spatial specifications, it remains a simplified representation of a considerably more complex phenomenon. A large share of the variability in municipal tourism demand remains unexplained, and the model relies on functional-form and distributional assumptions, such as log-linearity and a single, fixed spatial weights matrix, that may not fully capture the true underlying relationships.

Cross-sectional and static design. The analysis is based on a single cross-section of municipalities and therefore cannot capture how tourism demand, resource endowments and spatial spillovers evolve over time, leaving seasonal patterns, multi-year trends and the effects of specific shocks or events outside the scope of the current analysis.

4.3 Future work

Building on these limitations, several directions could extend and strengthen the present analysis.

First, extending the dataset to a panel structure covering multiple years would allow the persistence of spatial clusters and the stability of spillover effects to be assessed over time, and would help disentangle structural spatial relationships from transitory shocks to tourism demand.

Second, future work could complement the resource-based explanatory variables with flow-based measures of tourist mobility, such as mobile-network or geolocation data, which would allow the spatial spillover mechanisms identified here to be validated against actual patterns of visitor movement across neighbouring municipalities.

Finally, future research could disaggregate tourism demand by type of visitor (for example, domestic versus international, or leisure versus business tourism) and by accommodation type, which may respond differently to local resources and to the tourism endowment of neighbouring municipalities, thereby offering a more granular understanding of the complementarity and competition mechanisms identified in this report.

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